G11, Phy, U4:

Unit 4: Force, Work, Energy, and Power

The Concept of Force

A force is a push or a pull exerted on a body that changes its state of motion or causes deformation by changing its shape or size.

- **Vector Nature:** Force is a vector quantity determined by both magnitude and direction, represented as F or bold $\mathbf{F}$. It can be added using the head-to-tail method or trigonometric methods.
- **Measurement:** Forces are measured using the deformation of a spring. When a force is applied, the spring elongates, and a pointer on a calibrated scale reads the force value.
- **SI Unit:** The SI unit of force is the newton (N).

Classification of Forces

1. **Contact Forces:** Forces that involve direct physical contact between two bodies, with the force experienced only while contact is maintained.
   - *Examples:* Applied forces, normal forces, frictional forces, and spring forces.
2. **Non-Contact Forces:** Forces that act on a body without physical contact.
   - *Examples:* Magnetic force (e.g., attraction of iron to a magnet) and gravitational force (e.g., an object falling back to Earth).

Newton's First Law of Motion (Law of Inertia)

A body continues to be in its state of rest or of uniform motion in a straight line unless it is acted on by an unbalanced external force.

- **Inertia:** An inherent property of matter by which it continues in its existing state of rest or uniform motion in a straight line, resisting any attempt to change its velocity.
- **Mass as a Measure of Inertia:** Mass is the quantitative measure of an object's resistance to a change in its state of motion. Mass is constant everywhere. Objects with larger masses possess larger inertia and are more resistant to changes in velocity.
- **Inertial Frames of Reference:** A special set of reference frames in which Newton's first law is valid.
  - If an object does not interact with other objects, its acceleration is zero in an inertial frame.
  - Any reference frame moving with a constant velocity relative to an inertial frame is itself an inertial frame.

Newton's Second Law of Motion

The acceleration of a body is directly proportional to the net force acting on it and inversely proportional to the mass of the body.

The Net Force Formula

$$F = ma$$

- **Components:**
  - F: Net force acting on the body, measured in Newtons (N) or kilogram-metres per second squared ($\text{kg} \cdot \text{m/s}^2$).
  - m: Mass of the body, measured in kilograms (kg).
  - a: Acceleration of the body, measured in metres per second squared (m/s^2).
- **Application:** Used to calculate the acceleration produced when an unbalanced force acts on an object of a known mass. The direction of acceleration is always identical to the direction of the net force.
- **Definition of 1 Newton:** $1\ \text{N} = 1\ \text{kg} \cdot \text{m/s}^2$. It is the force that, acting on an object of mass 1 kg, produces an acceleration of $1\ \text{m/s}^2$.

Gravitational Force (Weight) Formula

$$W = mg \quad \text{or} \quad F_g = mg$$

- **Components:**
  - W (or F_g): Weight or gravitational force, measured in Newtons (N).
  - m: Mass of the body, measured in kilograms (kg).
  - g: Acceleration due to gravity, measured in metres per second squared (m/s^2).
- **Application:** Calculate the magnitude of the gravitational pull on an object directed toward the centre of the Earth.
- **Earth's Surface Value:** $g \approx 9.8\ \text{m/s}^2$ (or $10\ \text{m/s}^2$ in simplified scenarios).
- **Variability:** g decreases with height. On the Moon, $g_{\text{Moon}} \approx \frac{1}{6} g_{\text{Earth}}$.

Newton's Third Law of Motion

Every action has an equal and opposite reaction. Forces always act in pairs.

Action–Reaction Relationship

$$F_{12} = -F_{21}$$

- **Components:**
  - F_{12}: Force exerted by object 1 on object 2, in Newtons (N).
  - F_{21}: Force exerted by object 2 on object 1, in Newtons (N).
- **Application:** Analyze interacting bodies. Action and reaction are equal in magnitude, opposite in direction, and act on *different* objects — so they never cancel.
- **Example (Block on Table):** A block exerts a downward force on a table equal to its weight ($W = mg$); the table exerts an equal upward normal force on the block.

Forces of Friction

Frictional force (f) is the resistance to the motion of an object offered by a contacting surface.

- **Surface Dependency:** Magnitude depends on the nature of the surfaces (rougher ⇒ larger friction).
- **Normal Force Dependency:** Friction is directly proportional to the normal force: $f_k = \mu_k F_N$, $0 \le f_s \le \mu_s F_N$.

Types of Friction

1. **Static Friction:** Exists when a body tends to slide without moving; requires a relatively large force to overcome.
2. **Kinetic Friction:** Acts when surfaces slide or roll; its magnitude is less than maximum static friction for the same surfaces.

Work & Kinetic Energy

Work is done when an applied force causes a displacement parallel to the force. It is the mechanical transfer of energy.

Work Formula

$$W = FS\cos\theta$$

- **Components:** W in Joules (J); F in Newtons (N); S in metres (m); θ = angle between F and S.
- **Special cases:** $\theta = 0^\circ \Rightarrow W = FS$ (maximum); $\theta = 90^\circ \Rightarrow W = 0$; $\theta = 180^\circ \Rightarrow W = -FS$ (e.g. friction).

Kinetic Energy Formula

$$E_k = \tfrac{1}{2}mv^2$$

- **Components:** E_k in Joules (J); m in kg; v in m/s.
- **Application:** Motion-based energy of a moving body. Doubling v quadruples E_k.

Gravitational Potential Energy

Gravitational potential energy (E_p) is the stored energy possessed by an object due to its vertical position relative to a reference level, working against the downward pull of gravity.

Gravitational Potential Energy Formula

$$E_p = mgh$$

- **Components:**
  - E_p: Gravitational potential energy, measured in Joules (J).
  - m: Mass of the object, measured in kilograms (kg).
  - g: Acceleration due to gravity, in metres per second squared (m/s^2).
  - h: Vertical height above the reference level, measured in metres (m).
- **Application:** Applied to calculate stored energy based on height and mass.

Mechanical Energy

Mechanical energy is the total sum of an object's kinetic and potential energies:

$$E_{\text{mechanical}} = E_k + E_p$$

Power

Power is defined as the rate at which work is done or the rate at which energy is being transferred. It is a scalar quantity.

Power Formula

$$P = \frac{W}{t} \quad \text{or} \quad P = \frac{E}{t}$$

- **Components:**
  - P: Power, measured in Watts (W) or Joules per second (J/s).
  - W: Work done, measured in Joules (J). E: Energy transferred, in Joules (J).
  - t: Time taken for the transfer/work, measured in seconds (s).
- **Application:** Used to measure how rapidly work is performed or how quickly energy is consumed/produced by machines, motors, or pumps.
- **Definition of 1 Watt:** $1\ \text{W} = 1\ \text{J/s}$. It represents the power developed when 1 J of energy is transferred in 1 s.

Comprehensive Unit Synthesis

1. **Systematic Classification of Forces:** Mechanical systems are driven by forces classified into contact forces (requiring physical touch — friction, normal force, tension, spring forces) and non-contact forces (field forces acting through space — gravity, magnetism).
2. **Work–Energy Connection:** Work ($W = F_{\parallel} S$) is the mechanical process of translating forces into energy transfers, changing either kinetic energy ($E_k = \frac{1}{2}mv^2$) or potential energy ($E_p = mgh$).
3. **The Temporal Element (Power):** Performing the same work in half the time requires doubling the power output ($P = W/t$).

Formula index (table demo)

Formula	Use it when
$F = ma$	Net force / acceleration
$W = mg$	Weight of a mass
$E_k = \frac{1}{2}mv^2$	Energy of motion
$E_p = mgh$	Stored energy of height
$P = \frac{W}{t}$	Rate of doing work

KEY NOTES

1. **Zero Work Conditions:**
   1. If displacement is zero, work is zero — holding a heavy book still for two hours does $W = 0$.
   2. If force is perpendicular to displacement, work is zero — the lifting force does no work when carrying a bucket horizontally.
- **Friction Extremes:** $f_{\text{static}} > f_{\text{kinetic}}$ for the same surfaces.
- **Misconception:** Action–reaction pairs never cancel — they act on different bodies.
- **Unit Conversions:** Always convert mass to kg ($200\ \text{g} = 0.2\ \text{kg}$).
- **Links work too:** see the KaTeX supported functions reference.

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